

Example 1: Marketing Leads

****Problem.**** Campaign A ($n_1 = 18$) produced $\bar{x}_1 = 45$ leads,

$s_1 = 5.2$. Campaign B ($n_2 = 20$) produced $\bar{x}_2 = 41$ leads, $s_2 =$

6.0. Test $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq \mu_2$ at $\alpha = 0.05$.

****Solution.****

- Pooled $s = \sqrt{((17)(5.2^2) + (19)(6.0^2)) / 36} \approx 5.62$.
- $t = (45 - 41) / (5.62 \cdot \sqrt{(1/18 + 1/20)}) \approx 2.32$.
- $df = 36$, critical $t = \pm 2.028 \rightarrow |t| > t^* \Rightarrow$ reject H_0 .
- 95% CI: $(45 - 41) \pm 2.028 \cdot 5.62 \cdot \sqrt{(1/18 + 1/20)} \Rightarrow (1.0, 6.0)$

leads.

- $d = (45 - 41)/5.62 \approx 0.71$ (medium-large).
- Conclusion: Campaign A outperforms B.

Example 2: Diet Comparison (Right-Tailed)

****Problem.**** Diet 1 ($n_1 = 12$) mean loss 8.1 kg, $s_1 = 2.0$;

Diet 2 ($n_2 = 10$) mean loss 6.5 kg, $s_2 = 1.7$. Test $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 > \mu_2$ at $\alpha = 0.01$.

****Solution.****

- $s = \sqrt{((11)(2.0^2) + (9)(1.7^2)) / 20} \approx 1.87$.
- $t = (8.1 - 6.5) / (1.87 \cdot \sqrt{(1/12 + 1/10)}) \approx 2.44$.
- $df = 20$, right-tailed critical $t = 2.528 \Rightarrow t < \text{crit} \Rightarrow$ fail

to reject H_0 at 1%; evidence insufficient despite practical difference.

- 95% CI: $(1.6 \pm 2.086 \cdot 1.87 \cdot \sqrt{(1/12 + 1/10)}) \approx (-0.1, 3.3)$ kg

includes 0.

- $d \approx 0.86$.
- Conclusion: Not significant at 1%; would be significant at

$\alpha = 0.05$.